Concept Questions: Chapter 5

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ME-3640 – Heat Transfer

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Abstract

Fundamental principals and concepts are far more important to engineering expertise than understanding of example systems. These concept questions (shamelessly lifted from the textbook) will help you test your knowledge after reading the chapter summary and building a concept map. Use the relative ease and difficulty you experience while answering these questions to prioritize your efforts fo the remainder of this chapter.

1 Why Numerical Methods?

5–1C With powerful computers and software packages readily available, do you think obtaining analytical solutions to engineering problems will eventually disappear from engineering curricula?

5–2C What are the limitations of the analytical solution methods?

5–3C Consider a heat conduction problem that can be solved both analytically, by solving the governing differential equation and applying the boundary conditions, and numerically, by a software package available on your computer. Which approach would you use to solve this problem? Explain your reasoning.

5–4C What is the basis of the energy balance method? How does it differ from the formal finite difference method? For a specified nodal network, will these two methods result in the same or a different set of equations?

5–5C How do numerical solution methods differ from analytical ones? What are the advantages and disadvantages of numerical and analytical methods?

5–6C Two engineers are to solve an actual heat transfer problem in a manufacturing facility. Engineer A makes the necessary simplifying assumptions and solves the problem analytically, while engineer B solves it numerically using a powerful software package. Engineer A claims he solved the problem exactly and thus his results are better, while engineer B claims that he used a more realistic model and thus his results are better. To resolve the dispute, you are asked to solve the problem experimentally in a lab. Which engineer do you think the experiments will prove right? Explain.

2 Finite Difference Formulation of Differential Equations

5–7C Define these terms used in the finite difference formulation: node, nodal network, volume element, nodal spacing, and difference equation.

3 One-Dimensional Steady Heat Conduction

5–13C Explain how the finite difference form of a heat conduction problem is obtained by the energy balance method.

5–14C What are the basic steps involved in solving a system of equations with Gauss-Seidel method?

5–15C Consider a medium in which the finite difference formulation of a general interior node is given in its simplest form as

$$\frac{T_{m-1} - T_m + T_{m+1}}{\Delta x^2} + \frac{\dot{e}_m}{k} = 0 \tag{1}$$

1. Is heat transfer in this medium steady or transient?
2. Is heat transfer one-, two-, or three-dimensional?
3. Is there heat generation in the medium?
4. Is the nodal spacing constant or variable?
5. Is the thermal conductivity of the medium constant or variable?

5–16C How is an insulated boundary handled in finite difference formulation of a problem? How does a symmetry line differ from an insulated boundary in the finite difference formulation?

5–17C How can a node on an insulated boundary be treated as an interior node in the finite difference formulation of a plane wall? Explain.

5–18C In the energy balance formulation of the finite difference method, it is recommended that all heat transfer at the boundaries of the volume element be assumed to be into the volume element even for steady heat conduction. Is this a valid recommendation even though it seems to violate the conservation of energy principle?

4 Two-Dimensional Steady Heat Conduction

5–54C What is an irregular boundary? What is a practical way of handling irregular boundary surfaces with the finite difference method?

5–55C Consider a medium in which the finite difference formulation of a general interior node is given in its simplest form as the following:

$$T_{node} = \frac{T_{left} + T_{top} + T_{right} + T_{bottom}}{4} \tag{2}$$

1. Is heat transfer in this medium steady or transient?
2. Is heat transfer one-, two-, or three-dimensional?
3. Is there heat generation in the medium?
4. Is the nodal spacing constant or variable?
5. Is the thermal conductivity of the medium constant or variable?

5–56C Consider a medium in which the finite difference formulation of a general interior node is given in its simplest form as the following:

$$T_{left} + T_{top} + T_{right} + T_{bottom} - 4T_{node} + \frac{\dot{e}_{node} l^2}{k} = 0 \tag{3}$$

1. Is heat transfer in this medium steady or transient?
2. Is heat transfer one-, two-, or three-dimensional?
3. Is there heat generation in the medium?
4. Is the nodal spacing constant or variable?
5. Is the thermal conductivity of the medium constant or variable?

5 Transient Heat Conduction

5–85C How does the finite difference formulation of a transient heat conduction problem differ from that of a steady heat conduction problem? What does the term $\rho A\ \Delta x\ C_p \frac{T_m^{i+1} - T_m^i}{\Delta t}$ represent in the transient finite difference formulation?

5–86C What are the two basic methods of solution of transient problems based on finite differencing? How do heat transfer terms in the energy balance formulation differ in the two methods?

5–87C The explicit finite difference formulation of a general interior node for transient heat conduction in a plane wall is given by the following:

$$T_{m-1}^i - 2T_m^i + T_{m+1}^i + \frac{\dot{e}_m^i \Delta x^2}{k} = \frac{T_m^{i+1} - T_m^i}{\tau} \tag{4}$$

Obtain the finite difference formulation for the steady case by simplifying the relation above.

5–88C Consider transient one-dimensional heat conduction in a plane wall that is to be solved by the explicit method. If both sides of the wall are at specified temperatures, express the stability criterion for this problem in its simplest form.

5–89C Consider transient one-dimensional heat conduction in a plane wall that is to be solved by the explicit method. If both sides of the wall are subjected to specified heat flux, express the stability criterion for this problem in its simplest form.

5–90C The explicit finite difference formulation of a general interior node for transient two-dimensional heat conduction is given by the following:

$$T_{node}^{i+1} = \tau\left(T_{left}^{i} + T_{top}^{i} + T_{right}^{i} + T_{bottom}^{i}\right) + (1-4\tau)T_{node}^{i} + \tau\frac{\dot{e}_{node}^{i} l^2}{k} \tag{5}$$

Obtain the finite difference formulation for the steady case by simplifying the relation above.

5–91C Is there any limitation on the size of the time step Δt in the solution of transient heat conduction problems using each of the following methods:

1. the explicit method.
2. the implicit method.

5–92C Express the general stability criterion for the explicit method of solution of transient heat conduction problems.

5–93C Consider transient two-dimensional heat conduction in a rectangular region that is to be solved by the explicit method. If all boundaries of the region are either insulated or at specified temperatures, express the stability criterion for this problem in its simplest form.

5–94C The implicit method is unconditionally stable and thus any value of time step Δt can be used in the solution of transient heat conduction problems. To minimize the computation time, someone suggests using a very large value of Δt since there is no danger of instability. Do you agree with this suggestion? Explain.

6 Special Topic: Controlling the Numerical Error

5–121C Why do the results obtained using a numerical method differ from the exact results obtained analytically? What are the causes of this difference?

5–122C What is the cause of the discretization error? How does the global discretization error differ from the local discretization error?

5–123C Can the global (accumulated) discretization error be less than the local error during a step? Explain.

5–124C How is the finite difference formulation for the first derivative related to the Taylor series expansion of the solution function?

5–125C Explain why the local discretization error of the finite difference method is proportional to the square of the step size. Also explain why the global discretization error is proportional to the step size itself.

5–126C What causes the round-off error? What kind of calculations are most susceptible to round-off error?

5–127C What happens to the discretization and the round-off errors as the step size is decreased?

5–128C Suggest some practical ways of reducing the round-off error.

5–129C What is a practical way of checking if the round-off error has been significant in calculations?

5–130C What is a practical way of checking if the discretization error has been significant in calculations?

A Rubrics

Submissions of this assignment will be graded upon the criteria found in this section. Review these carefully, and complete your work so that application of this rubric produces your desired score.

Criterion 1 Problems

Homework is completed with appropriate effort, neatness, etc.

Points	Description
20	Honest effort to participate and learn: every problem shows an answer or questions which need resolution.
10	Rushed or haphazard work: some problems show no effort at all.
0	Work is absent, wholly incorrect, or plagiarized.